EXCELERATE DATA VISUALIZATION ASSOCIATE INTERNSHIP

Learner & Opportunity Analytics

Final Presentation & Program Closure

Team 34

Sudheer Muthyala • Simpy Mandal • Krishna Swamy Tejesh Kumar

Datasets: Opportunity Data | Learner Data

OVERVIEW

Agenda

01

Project Objective

Why this analysis, and what it supports

02

Datasets

Learner & Opportunity data at a glance

03

Analytical Pipeline

Setup → cleaning → integration → dashboards

04

Data Quality

Issues found and how they were resolved

05

Key Findings

Category, payment, workflow & engagement trends

06

Insights & Recommendations

What the data means for the business

01 · PROJECT OBJECTIVE

From Raw Data to Decision-Ready Dashboards

Team 34 was tasked with turning two raw platform exports — Learner and Opportunity records — into a clean, integrated dataset and a set of dashboards that let Excelerate stakeholders understand applications, completions, and engagement across opportunity types.

- Stand up a cloud PostgreSQL environment and connect it to Looker Studio
- Clean and validate the Learner and Opportunity datasets
- Integrate both datasets into a single analytical table
- Surface KPIs and trends through interactive dashboards
- Translate findings into evidence-based recommendations

PIPELINE AT A GLANCE

15,029 → 14,793

Learner records, after cleaning

5,733 → 5,728

Opportunity records, after cleaning

14,791

Integrated learner–opportunity records

3

Weeks of setup, cleaning & dashboarding

Two Datasets, One Shared Identifier

Learner Dataset

learner_clean · Key: sk

14,793 records (from 15,029 raw; 236 corrupted rows removed)

- Application ID, status & assigned cohort
- Apply date, completion date, timestamps
- Payment status (paid / unpaid)
- 43 total attributes captured

Cleaning: removed embedded PDF/Base64 records, standardized "XploreU" category values to NULL, validated payment & workflow status.

Opportunity Dataset

opportunity_clean · Key: opportunity_id

5,728 records (from 5,733 raw; 5 corrupted rows removed)

- Category, badge, duration & fee
- Location, eligibility & rewards
- Auto-approval & archive flags
- 33 total attributes captured

Cleaning: standardized location values (Virtual / Work From Home), replaced invalid categories with NULL, converted text "NULL" strings to SQL NULL.

From Raw Exports to Live Dashboards

1

Database Setup

PostgreSQL hosted on Supabase; connected to Looker Studio via the PostgreSQL connector

2

Data Cleaning

Removed corrupted rows, standardized categories & locations, converted blanks to SQL NULL

3

Integration

Joined Learner (sk) and Opportunity (opportunity_id) on a normalized shared identifier

4

Validation

Verified record counts, duplicate keys, and missing-value patterns before reporting

5

Dashboarding

Built KPI cards and charts in Looker Studio for ongoing monitoring

Before vs. After Cleaning

Metric	Learner	Opportunity
Raw records	15,029	5,733
Corrupted / invalid records removed	236 (embedded PDF/Base64)	5 (malformed IDs & testimonials)
Final clean records	14,793	5,728
Invalid category values	“XploreU” → SQL NULL	“XploreU” → SQL NULL
Location / text standardization	N/A	Virtual, WFH variants → 2 consistent labels
Duplicate key check	Duplicate pk values found (Week 1) — resolved during cleaning	No duplicate opportunity_id after cleaning

03 · INTEGRATION

Joining Learner and Opportunity Records

No shared column existed at first glance. Investigation revealed the Learner table's sk field and the Opportunity table's opportunity_id field carried the same underlying ID under different prefixes.

```
Learner#0000000010Z56C78JF7M6VQXQ2
Opportunity#0000000010Z56C78JF7M6VQXQ2
```

Prefixes stripped via PostgreSQL REPLACE(), then joined with an INNER JOIN on the normalized ID.

Resulting table: learner_opportunity

- Learner application & payment details
- Opportunity category, location, fee, duration & reward
- Application and opportunity timestamps

INTEGRATION RESULT

14,791

matched learner-opportunity records

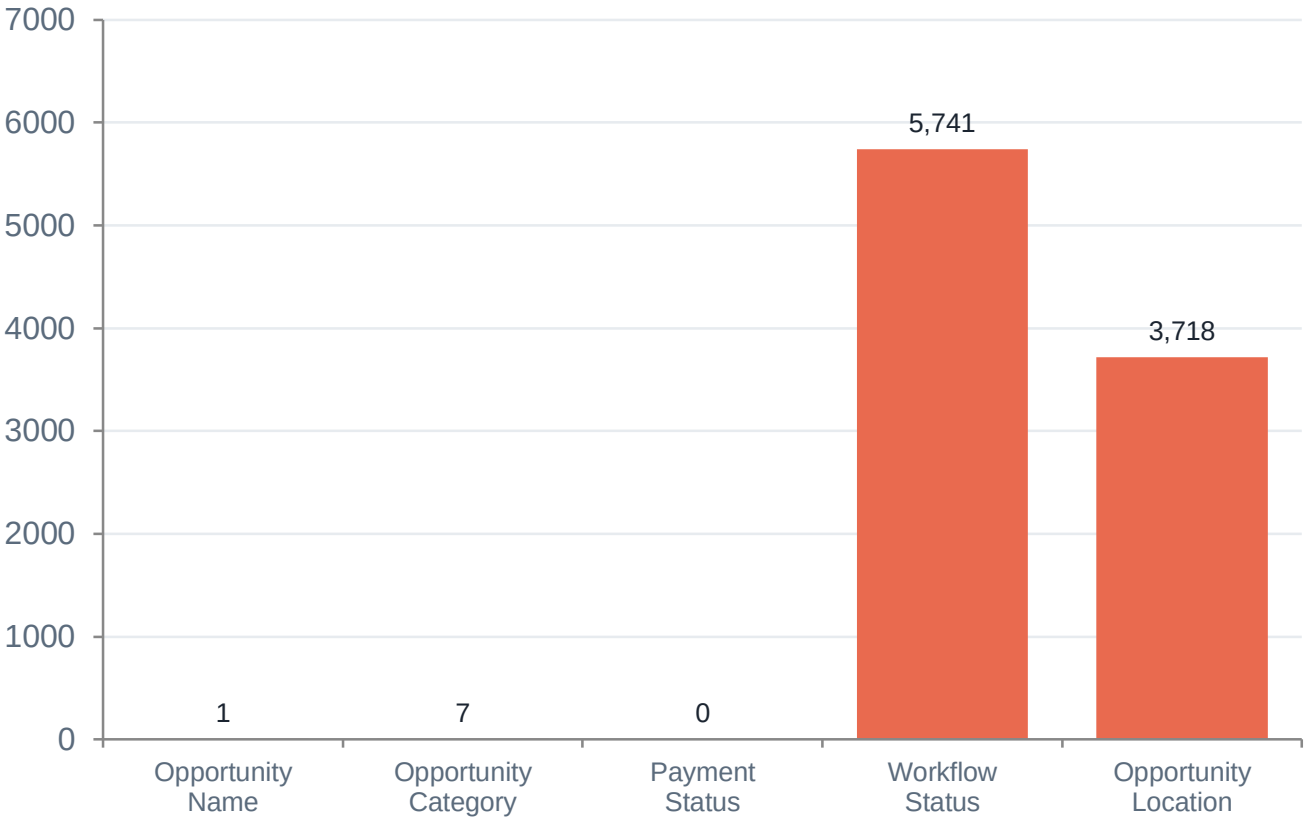
Duplicate column names (pk, category, fee, created_at, modified_at) renamed to avoid ambiguity

Blank strings in status and opportunity_location converted to SQL NULL

Source anomalies preserved (not overwritten) to protect data integrity

Remaining Missing Values — Understood, Not Ignored

After cleaning and integration, a final validation pass quantified remaining gaps. Each was traced to a genuine source-data limitation rather than a processing error, so values were preserved as SQL NULL instead of guessed or defaulted.



Workflow Status — 5,741

Not captured for records still awaiting a workflow update in the source system.

Opportunity Location — 3,718

Location left blank in the source opportunity records; not inferred.

Opportunity Category — 7

Invalid “XploreU” values intentionally converted to NULL.

Opportunity Name / Payment — 1 / 0

A single source gap; payment status is fully populated.

The Integrated Dataset at a Glance

14,791

Total applications

learner × opportunity matches

5,728

Opportunities represented

across 8 categories

6,076

Work-From-Home applications

the single largest location segment

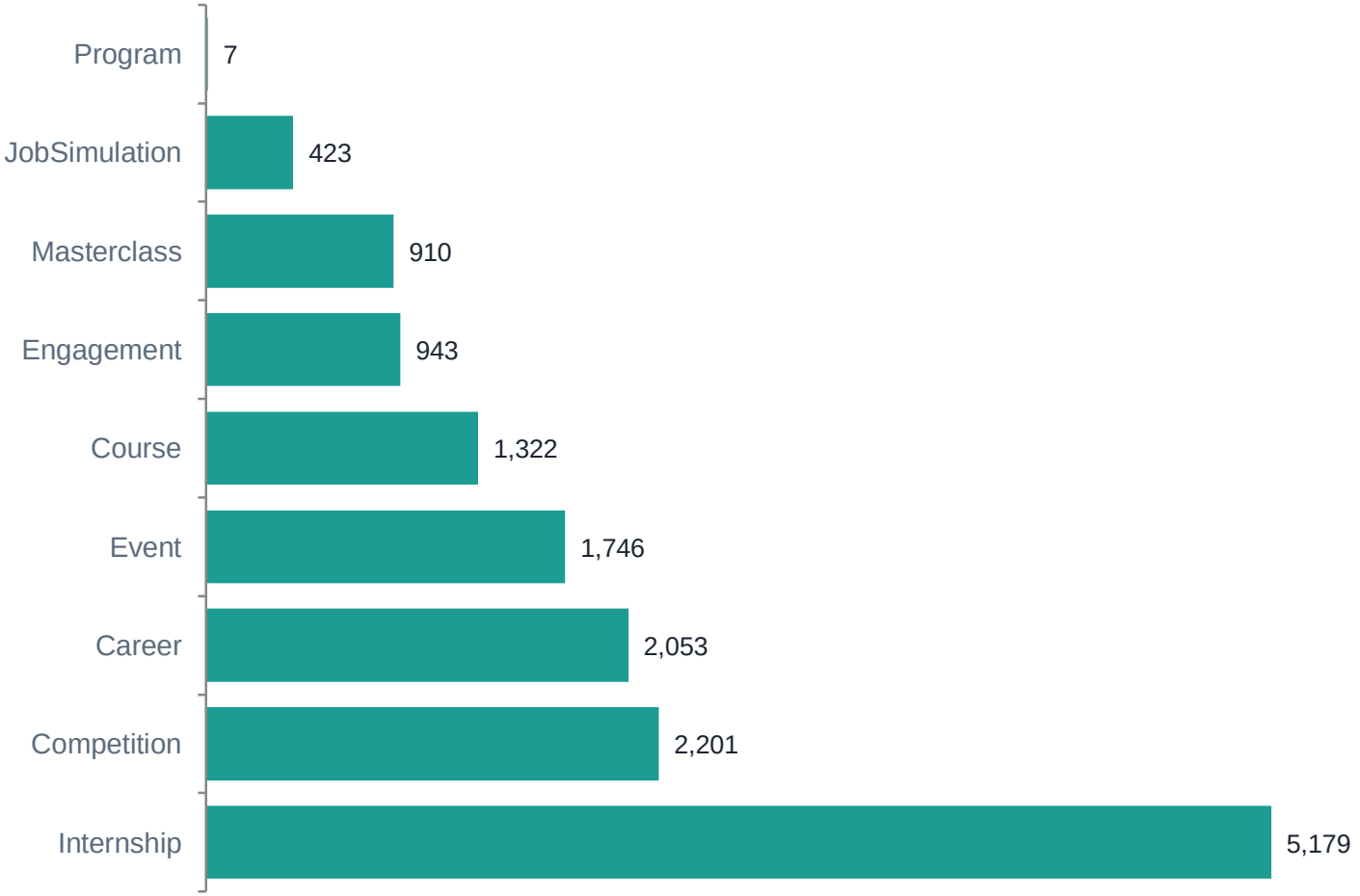
9,650

Applications to week-long

opportunities — the most common duration

Source: learner_opportunity integrated table (SQL validation, Week 2–3).

Applications by Opportunity Category



OBSERVATION

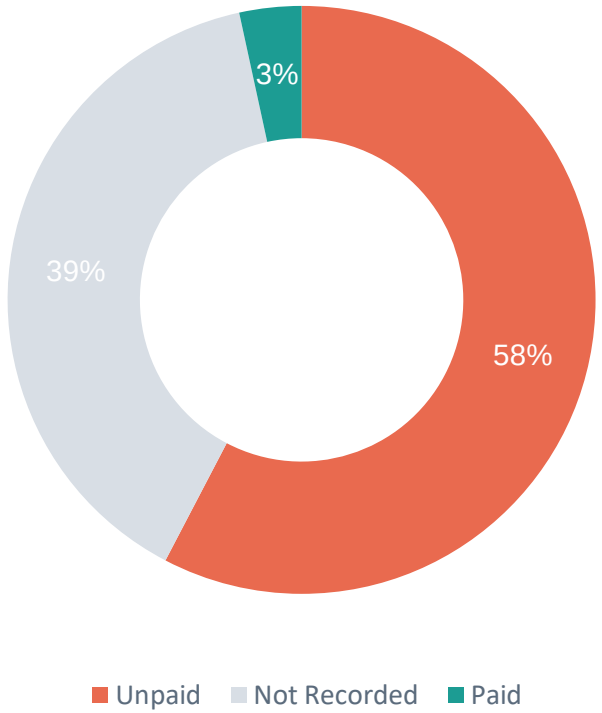
Internships dominate demand, drawing 5,179 applications — more than double the next category (Competitions, 2,201).

BUSINESS INSIGHT

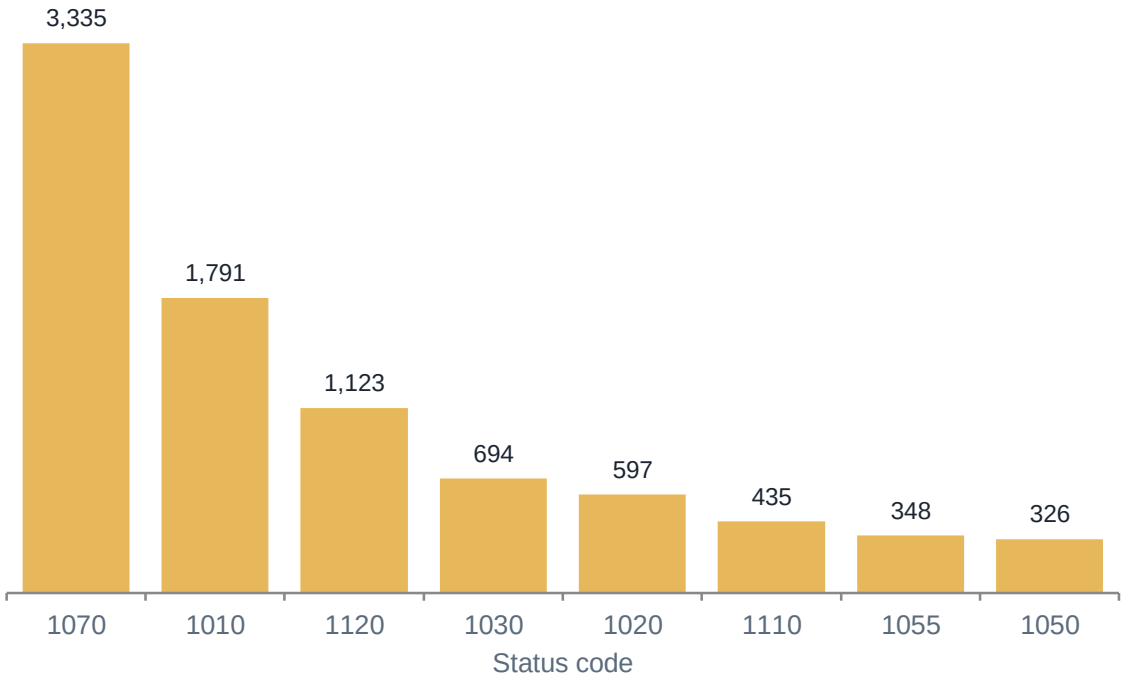
Internship listings should be prioritized for supply and marketing investment; Masterclass and JobSimulation categories have room to grow through targeted promotion.

Payment & Workflow Status

Payment Status



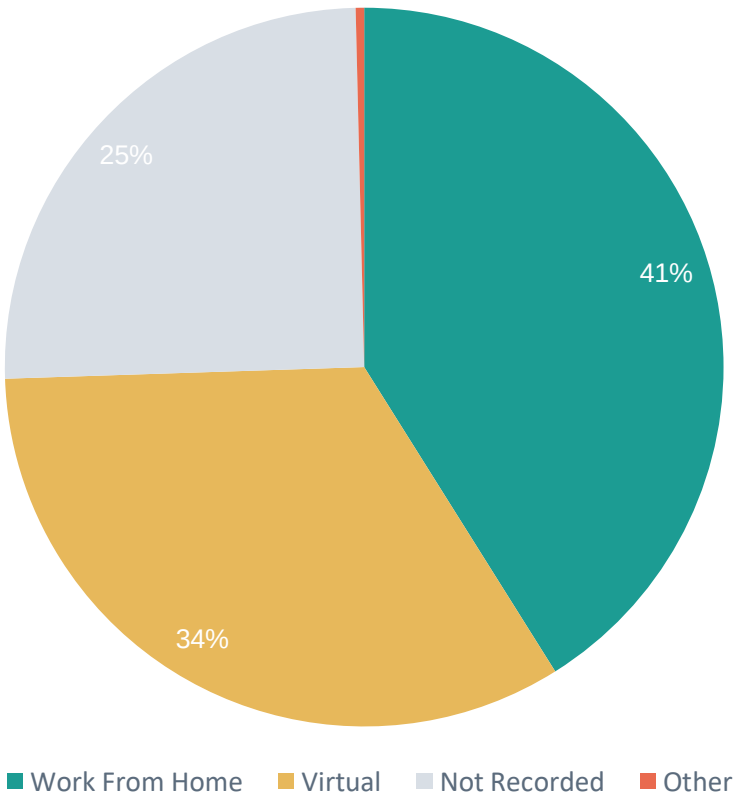
Workflow Status (Top Stages)



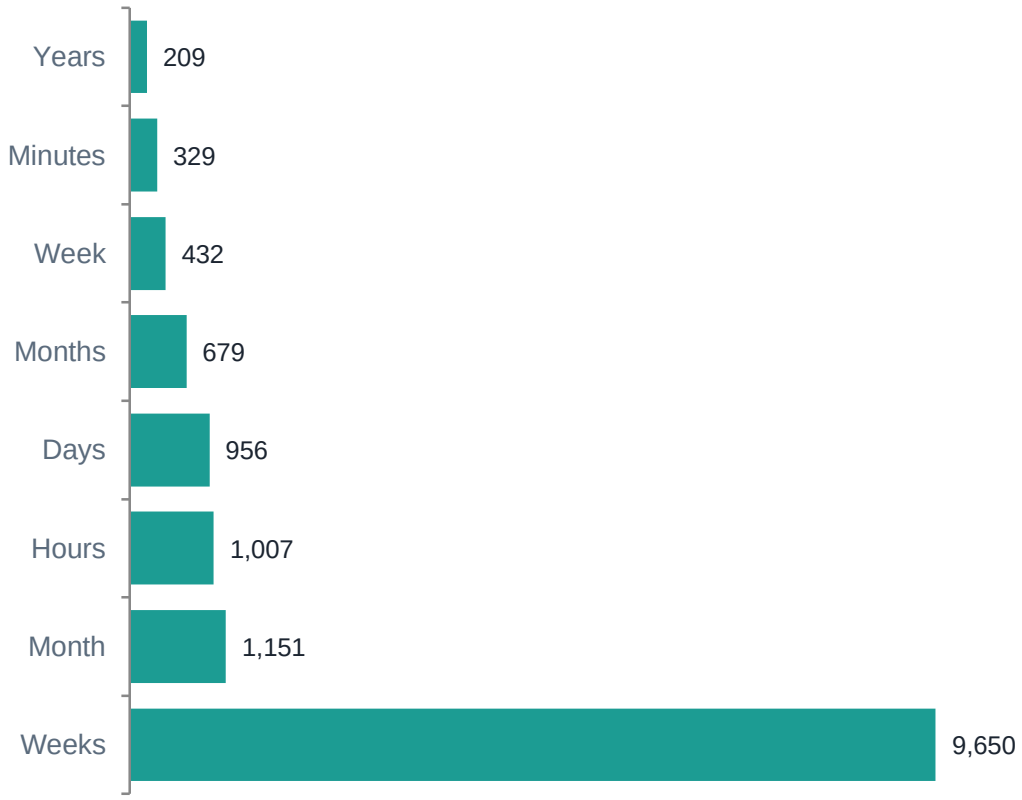
94% of applications with a recorded payment status are unpaid — most opportunities are free to join. Status code 1070 accounts for the largest single workflow stage (3,335 records); status codes are numeric platform stages rather than descriptive labels.

Where and How Long: Location & Duration

Opportunity Location



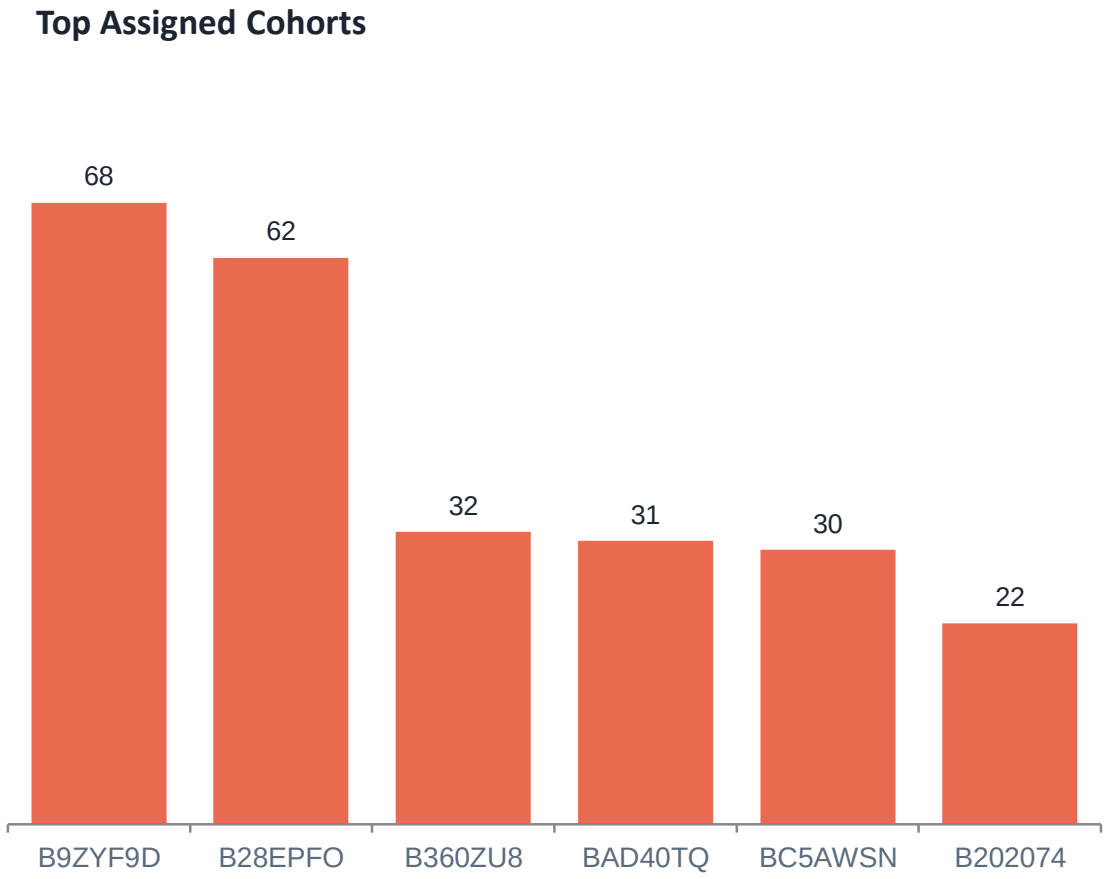
Opportunity Duration Type



Most Applied Opportunities & Cohorts

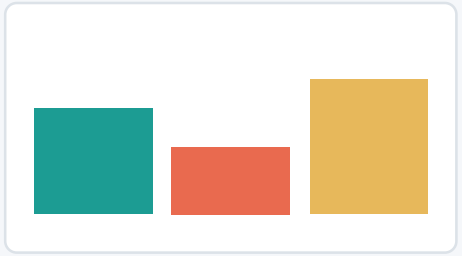
Trainee Internship	119
Bulk Users Applied to Internship	69
Competition with 4 Members	43
developement121	42
Event Automation 1691239404701	38
Mini Saga 3.0	36

Note: 9,239 applications carry no assigned cohort in the source data — excluded here to surface active cohort activity.



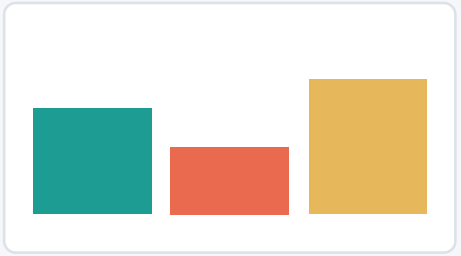
Looker Studio Dashboard Design

Findings were operationalized into a Looker Studio dashboard connected live to the Supabase PostgreSQL database, giving stakeholders a filterable, always-current view of applications, sign-ups, completions, and rewards.



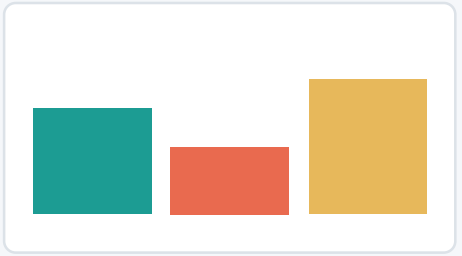
KPI Cards

Total applications, sign-ups, completion rate & rewards at a glance



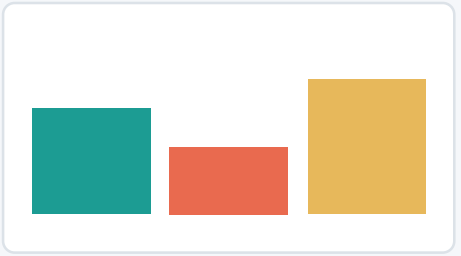
Category & Trend Charts

Applications by opportunity category, and month-over-month trend



Filters

Slice by opportunity category, lead source, scholarship type & date range



Reward & Source Views

Reward-type distribution and lead-source performance

06 · BUSINESS INSIGHTS

What the Data Tells Us

Internships drive engagement

5,179 applications — nearly 2.5× the next category — make internships the platform's core product.

Remote-first participation

Work From Home (41%) and Virtual (33%) options together account for the vast majority of located applications.

Short-format learning wins

Week-long opportunities (9,650 applications) dominate over month- or year-long formats.

Most opportunities are free

94% of recorded payments are “unpaid,” suggesting free access is the primary growth driver.

Workflow visibility gap

5,741 applications lack a recorded workflow status — the single largest data-quality gap to close.

Concentrated cohort activity

A small set of cohorts (e.g., B9ZYF9D, B28EPFO) account for disproportionate application volume.

06 · RECOMMENDATIONS

Data-Backed Next Steps

Expand internship supply

Given 5,179 applications against 5,728 total opportunities, prioritize sourcing and promoting more internship listings to meet demonstrated demand.

Close the workflow-status gap

Require or default a workflow status at application intake to eliminate the 5,741 missing records and enable accurate funnel reporting.

Grow low-adoption categories

Masterclass and JobSimulation categories (under 1,000 applications each) have headroom — test targeted campaigns to lift participation.

Standardize location capture

3,718 applications have no recorded location; enforce a required, standardized location field for new opportunity listings.

Double down on short formats

Week-long formats outperform longer ones nearly 6× over — favor short, high-completion opportunity designs going forward.

Investigate cohort concentration

A handful of cohorts drive outsized volume; study what makes them successful and replicate that playbook elsewhere.

CONCLUSION

A Reliable, Analysis-Ready Foundation

Over the internship, Team 34 stood up a cloud database, cleaned and validated two real-world datasets, integrated them into a single 14,791-record analytical table, and translated the results into dashboards and evidence-based recommendations — a foundation Excelerate can build on for ongoing reporting.

Thank you.

Team 34 · Sudheer Muthyala, Simpy Mandal, Krishna Swamy Tejesh Kumar

Excelerate Data Visualization Associate Internship